

AI Knowledge Base Sample Document

Section 1

Artificial Intelligence: Artificial Intelligence (AI) enables computers to perform tasks that normally require human intelligence. Common applications include recommendation systems, language translation, fraud detection, and virtual assistants.

Machine Learning: Machine Learning is a subset of AI where models learn patterns from data. Supervised learning uses labeled data, unsupervised learning finds hidden patterns, and reinforcement learning learns through rewards.

Data Engineering: Data Engineering focuses on collecting, transforming, validating, and storing data. Typical tools include Python, SQL, Apache Spark, Airflow, Kafka, PostgreSQL, and cloud storage.

Cloud Computing: Cloud platforms such as AWS, Azure, and Google Cloud provide scalable compute, storage, networking, databases, and AI services.

Cybersecurity: Security best practices include least privilege access, encryption, MFA, regular backups, vulnerability scanning, and audit logging.

SQL: SQL is used to query relational databases. Frequently used commands include SELECT, INSERT, UPDATE, DELETE, JOIN, GROUP BY, HAVING, and window functions.

Python: Python is widely used for automation, web development, data analysis, AI, and scripting because of its readability and extensive ecosystem.

Technology	Primary Use	Example Tool
Programming	Application Development	Python
Database	Data Storage	PostgreSQL
Big Data	Distributed Processing	Apache Spark
Workflow	Scheduling	Airflow
Messaging	Streaming	Kafka
Cloud	Infrastructure	AWS

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